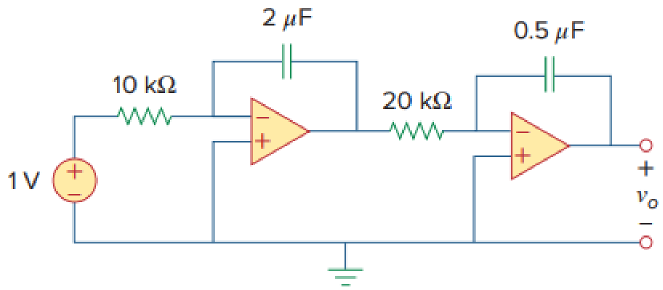
Problem

At t = 1.5 ms, calculate vo due to the cascaded integrators in Fig. 6.89. Assume that the integrators are reset to 0 V at t = 0.

Fig. 6.89:



Step-by-step solution

Step 1 of 1

The output of the first Op- amp is

$$V_1 = \frac{-1}{RC}\int V_i dt$$

$$V_1 = -\frac{1}{10\times10^3\times2\times10^{-6}}\int_0^t V_i dt$$

$$V_1 = \frac{-100t}{2}$$

$$V_1 = -50t$$

$$V_0 = \frac{-1}{RC}\int V_1 dt$$

$$V_0 = -\frac{1}{20\times10^3\times0.5\times10^{-6}}\int_0^t(-50t)dt$$

$$V_0 = 2500t^2$$

At $t = 1.5\,ms$

$$V_0 = 2500(1.5)^2\times10^{-6}$$

$$\therefore \boxed{V_0 = 5.625\,\text{mV}}$$